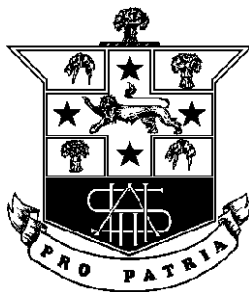


# HURLSTONE AGRICULTURAL HIGH SCHOOL



## MATHEMATICS – EXTENSION TWO

### 2006 HSC

#### ASSESSMENT TASK 1

**Examiners ~ G Rawson, Z Pethers**

#### **GENERAL INSTRUCTIONS**

- |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Reading Time – 3 minutes.</li><li>• Working Time – 40 MINUTES.</li><li>• Attempt <b>all</b> questions.</li><li>• <b>All</b> necessary working should be shown in every question.</li><li>• This paper contains two (2) questions.</li></ul> | <ul style="list-style-type: none"><li>• Marks may not be awarded for careless or badly arranged work.</li><li>• Board approved calculators may be used.</li><li>• <b>Each question is to be started on a new piece of paper.</b></li><li>• This examination paper must <b>NOT</b> be removed from the examination room.</li></ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

STUDENT NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

**QUESTION ONE**    **20 marks**    *Start a SEPARATE sheet*

(a) If  $z = 2 - i$  and  $w = 1 + 2i$ , find

- |       |               |          |
|-------|---------------|----------|
| (i)   | $z + w$       | <b>1</b> |
| (ii)  | $w - z$       | <b>1</b> |
| (iii) | $zw$          | <b>1</b> |
| (iv)  | $z\bar{w}$    | <b>1</b> |
| (v)   | $\frac{z}{w}$ | <b>2</b> |

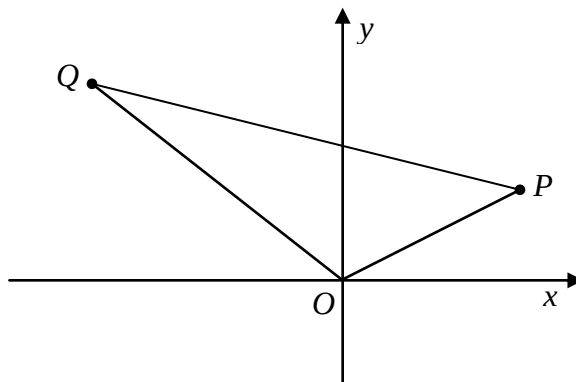
(b) Find all pairs of integers  $x$  and  $y$  that satisfy  $(x + iy)^2 = 24 + 10i$  **3**

(c) Consider the equation  $z^2 + az + (1 + i) = 0$ .  
Find the complex number  $a$ , given that  $i$  is a root of the equation **2**

(d) It is given that  $2 + i$  is a root of  $P(z) = z^3 + rz^2 + sz + 20$ , where  $r$  and  $s$  are real numbers

- |      |                                            |          |
|------|--------------------------------------------|----------|
| (i)  | State why $2 - i$ is also a root of $P(z)$ | <b>1</b> |
| (ii) | Factorise $P(z)$ over the real numbers     | <b>2</b> |

(e) The diagram below show a complex plane with origin  $O$ .  
The points  $P$  and  $Q$  represent arbitrary non-zero complex numbers  $z$  and  $w$  respectively.  
Thus the length of  $PQ$  is  $|z - w|$



- |       |                                                                                          |          |
|-------|------------------------------------------------------------------------------------------|----------|
| (i)   | Copy the diagram, and use it to show that $ z - w  \leq  z  +  w $                       | <b>2</b> |
| (ii)  | Construct the point $R$ representing $z + w$ .<br>What type of quadrilateral is $OPRQ$ ? | <b>2</b> |
| (iii) | If $ z - w  =  z + w $ , what can be said about the complex number $\frac{w}{z}$ ?       | <b>2</b> |

**QUESTION TWO**    **20 marks**    *Start a SEPARATE sheet*

(a) Given  $z = \sqrt{6} - \sqrt{2}i$ , find

- |       |                            |          |
|-------|----------------------------|----------|
| (i)   | $\operatorname{Re}(z^2)$   | <b>1</b> |
| (ii)  | $(\operatorname{Im} z)^2$  | <b>1</b> |
| (iii) | $ z $                      | <b>1</b> |
| (iv)  | $\arg z$                   | <b>2</b> |
| (v)   | $z^4$ in the form $x + iy$ | <b>2</b> |

(b) Let  $\alpha = 1 + \sqrt{3}i$  and  $\beta = 1 + i$

- |       |                                                                                                                                                                                        |          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (i)   | Find $\frac{\alpha}{\beta}$ in the form $x + iy$                                                                                                                                       | <b>2</b> |
| (ii)  | Express $\alpha$ in modulus-argument form                                                                                                                                              | <b>2</b> |
| (iii) | Given that $\beta$ has modulus-argument form $\beta = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$ ,<br>Find the modulus-argument form of $\frac{\alpha}{\beta}$ | <b>2</b> |
| (iv)  | Hence, find the exact value of $\sin \frac{\pi}{12}$                                                                                                                                   | <b>2</b> |

(c) Sketch the region in the complex plane where the inequalities

$$|z - 1 - i| < 2 \quad \text{and} \quad 0 < \arg(z - 1 - i) < \frac{\pi}{4}$$

hold simultaneously.

**3**

(d) The diagram below shows the locus of points  $z$  in the complex plane such that

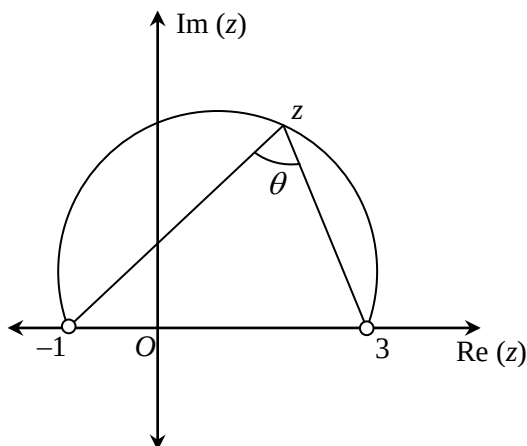
$$\arg(z - 3) - \arg(z + 1) = \frac{\pi}{3}.$$

This locus is part of a circle.

The angle between the lines from  $-1$  to  $z$  and from  $3$  to  $z$  is  $\theta$ , as shown.

Explain why  $\theta = \frac{\pi}{3}$

**2**



Mathematics Ext 2 Assessment Task 1 (2006 HSC) – Q1

(a)  $z = 2 - i, w = 1 + 2i$

$$(i) \quad \begin{aligned} z + w &= 2 - i + 1 + 2i \\ &= 3 + i \end{aligned}$$

$$(ii) \quad \begin{aligned} w - z &= (1 + 2i) - (2 - i) \\ &= -1 + 3i \end{aligned}$$

$$\begin{aligned} zw &= (2 - i)(1 + 2i) \\ (iii) \quad &= 2 + 4i - i - 2i^2 \\ &= 4 + 3i \end{aligned}$$

$$\begin{aligned} z\bar{w} &= (2 - i)\overline{(1 + 2i)} \\ (iv) \quad &= (2 - i)(1 - 2i) \\ &= 2 - 4i - i + 2i^2 \\ &= -5i \end{aligned}$$

$$\begin{aligned} \frac{z}{w} &= \frac{2 - i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i} \\ (v) \quad &= \frac{-5i}{1 + 4} \\ &= -i \end{aligned}$$

$$(b) \quad \begin{aligned} (x + iy)^2 &= 24 + 10i \\ x^2 - y^2 + 2xyi &= 24 + 10i \end{aligned}$$

$$\therefore \begin{cases} x^2 - y^2 = 24 & \dots(1) \\ 2xy = 10 & \dots(2) \end{cases}$$

**Method 1:**

$x, y$  must be integers, so by inspection,  $x = 5, y = 1$  or  $x = -5, y = -1$

**Method 2:**

Substituting  $y = \frac{5}{x}$  from (2) into (1):

$$\begin{aligned} x^2 - \frac{25}{x^2} - 24 &= 0 \\ x^4 - 24x^2 - 25 &= 0 \\ (x^2 - 25)(x^2 + 1) &= 0 \\ x &= \pm 5 \dots\dots (x \text{ is real}) \\ \therefore x &= 5, x = 1 \text{ or } x = -5, y = -1 \end{aligned}$$

(c)  $z^2 + az + (1 + i) = 0$

Substitute  $z = i$ , since  $i$  is a root

$$i^2 + ai + 1 + i = 0$$

$$-1 + ai + 1 + i = 0$$

$$i(a + 1) = 0$$

$$a = -1$$

- (d) (i) Since all the coefficients of  $P(z)$  are real, complex roots occur in conjugate pairs. So if  $2 + i$  is a root,  $2 - i$  must also be a root.

(ii) **Method 1:**

Let the other root be  $\alpha$ , and using product of roots...

$$(2 + i)(2 - i)\alpha = -\frac{20}{1}$$

$$(4 - i^2)\alpha = -20$$

$$5\alpha = -20$$

$$\alpha = -4$$

$$\therefore P(z) = [z - (-4)][z - (2 + i)][z - (2 - i)]$$

$$= (z + 4)(z^2 - 4z + 5)$$

**Method 2:**

Combining the known factors:

$$(z - 2 - i)(z - 2 + i) = (z - 2)^2 + 1 = z^2 - 4z + 5$$

$$\therefore P(z) = (z^2 - 4z + 5)(z + 4)$$

where the factor  $z + 4$  has been found by equating the coefficient of  $z^3$  and the constant.

- (e) (i) The sum of any two sides of a triangle is together greater than the third side  
so  $PQ < OQ + OP \quad \therefore |z - w| < |w| + |z| \quad \dots \textcircled{1}$

If  $POQ$  is straight,

$$\text{then } PQ = OQ + OP \quad \therefore |z - w| = |w| + |z| \quad \dots \textcircled{2}$$

From  $\textcircled{1}$  and  $\textcircled{2} \quad |z - w| \leq |w| + |z| \quad \underline{\text{NOTE:}}$  Must explain the  $<$  and the  $=$ .

- (ii) Parallelogram

- (iii)  $|z - w| = |z + w|$  means diagonals are  $=$ , so rectangle.

$$\text{Now, } \arg \frac{w}{z} = \arg w - \arg z = \angle QOP = \frac{\pi}{2} \text{ (rectangle)}$$

$$\therefore \frac{w}{z} \text{ is imaginary}$$

Mathematics Ext 2 Assessment Task 1 (2006 HSC) – Q2

(a)  $z = \sqrt{6} - \sqrt{2}i$

$$\begin{aligned} z^2 &= (\sqrt{6} - \sqrt{2}i)(\sqrt{6} - \sqrt{2}i) \\ &= 6 - 2 - 2\sqrt{12}i \\ &= 4 - 4\sqrt{3}i \\ \text{Re}(z^2) &= 4 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |z| &= \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} \\ &= 2\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \arg z &= \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{6}}\right) \\ &= \tan^{-1}(-\sqrt{3}) \\ &= -\frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} z^4 &= (z^2)^2 = (4 - 4\sqrt{3}i)^2 \\ \text{(iv)} \quad &= 16 - 48 - 32\sqrt{3}i \\ &= -32 - 32\sqrt{3}i \end{aligned} \quad \text{or} \quad \begin{aligned} z &= 2\sqrt{2}\text{cis}\left(-\frac{\pi}{6}\right) \\ z^4 &= (2\sqrt{2})^4 \text{cis}\left(-\frac{4\pi}{6}\right) \\ &= 64\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \end{aligned}$$

Marks: (i) correct answer – **1 mark**

(ii) correct answer – **1 mark**

(iii) correct answer – **2 marks**; correct method with incorrect values – **1 mark** only

(iv) correct answer – **2 marks**; correct method with incorrect values – **1 mark** only

$$\begin{aligned} \text{(b)} \quad \text{(i)} \quad &\frac{1+\sqrt{3}i}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{1-i+\sqrt{3}i+\sqrt{3}}{1+1} \\ &= \frac{1+\sqrt{3}}{2} + i\frac{\sqrt{3}-1}{2} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |\alpha| &= \sqrt{1^2 + (\sqrt{3})^2} = 2 \\ \arg \alpha &= \tan^{-1} \frac{\sqrt{3}}{1} = \frac{\pi}{3} \\ \therefore \alpha &= 2\text{cis}\frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{\alpha}{\beta} &= \frac{2\text{cis}\frac{\pi}{3}}{\sqrt{2}\text{cis}\frac{\pi}{4}} \\ &= \frac{2}{\sqrt{2}}\text{cis}\left(\frac{\pi}{3} - \frac{\pi}{4}\right) \\ &= \sqrt{2}\text{cis}\frac{\pi}{12} \end{aligned}$$

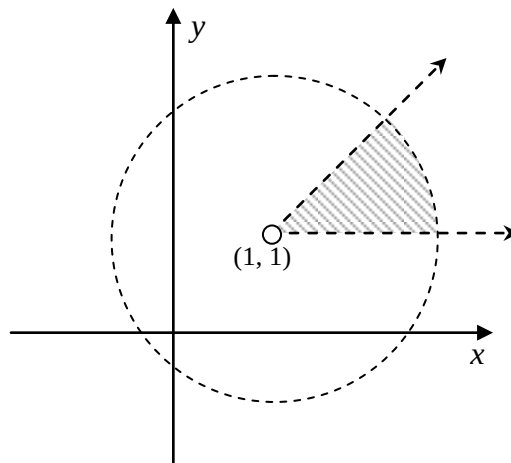
From (i) and (iii),

$$\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) = \frac{1+\sqrt{3}}{2} + i\frac{\sqrt{3}-1}{2}$$

(iv) so,  $\sqrt{2}\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2}$

$$\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

(c)



(d) Let  $\arg(z - 3) = \alpha$  and  $\arg(z + 1) = \beta$   
 $\alpha - \beta = \theta$  since the exterior angle of a triangle  
 is equal to the sum of the interior  
 opposite angles.

But we are told  $\alpha - \beta = \frac{\pi}{3}$ ,

$$\therefore \theta = \frac{\pi}{3}$$

